

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.6085045236342096, median 0.464548510997314, std: 0.5160939857871177
Gyroscope error (imu0): mean 0.6251483383419513, median 0.5515027010924173, std: 0.3618673255800177
Accelerometer error (imu0): mean 0.6954363126842715, median 0.6346219485380425, std: 0.3972926845218351

Residuals

Reprojection error (cam0) [px]: mean 0.6085045236342096, median 0.464548510997314, std: 0.5160939857871177
Gyroscope error (imu0) [rad/s]: mean 0.017541621509591522, median 0.015475129742388965, std: 0.010153973501470995
Accelerometer error (imu0) [m/s^2]: mean 0.09125694440044545, median 0.0832767268788441, std: 0.052133769492380445

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ 0.99987512  0.01520848  0.00429518  0.0225262 ]  
 [-0.01520471  0.99988399 -0.00090895 -0.00213086]  
 [-0.0043085  0.00084353  0.99999036 -0.00386642]  
 [ 0.         0.         0.         1.        ]]
```

T_ic: (cam0 to imu0):

```
[[ 0.99987512 -0.01520471 -0.0043085 -0.02257245]  
 [ 0.01520848  0.99988399  0.00084353  0.00179128]  
 [ 0.00429518 -0.00090895  0.99999036  0.00376769]  
 [ 0.         0.         0.         1.        ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.01334857234167497

Gravity vector in target coords: [m/s^2]

[-0.02192112 -0.03982915 -9.80644462]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [599.1739928442613, 599.2682460042912]

Principal point: [324.9719463276537, 250.51904788461366]

Distortion model: radtan

Distortion coefficients: [0.11030606589572783, -0.20736097595170666, -0.002276220265004869, 0.00184355291590457]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.08 [m]

Spacing 0.024 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 200

Accelerometer:

Noise density: 0.009278837334054291

Noise density (discrete): 0.1312225760087339

Random walk: 0.0002804825905759594

Gyroscope:

Noise density: 0.0019841370058405537

Noise density (discrete): 0.02805993463266056

Random walk: 1.7544399415266354e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

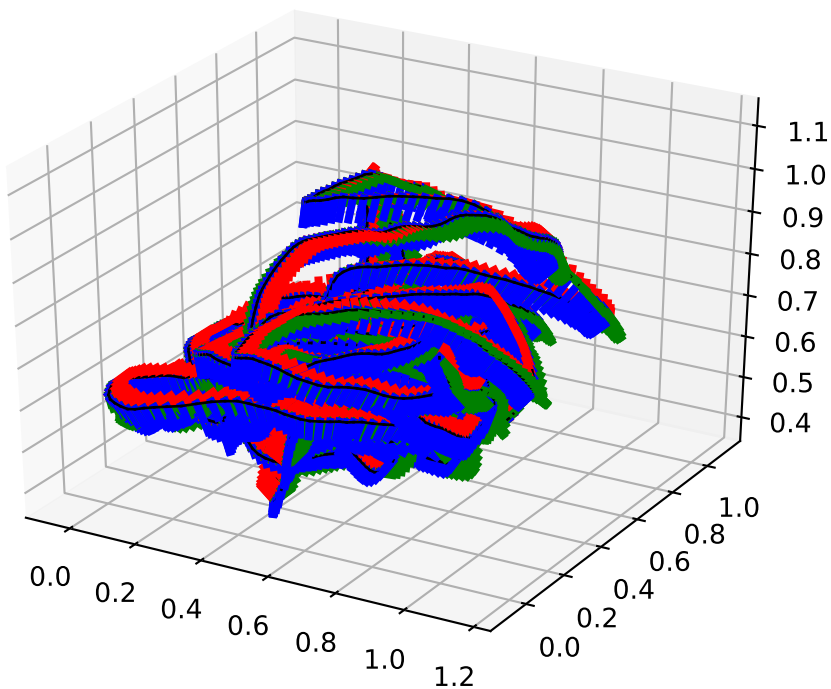
[0. 1. 0. 0.]

[0. 0. 1. 0.]

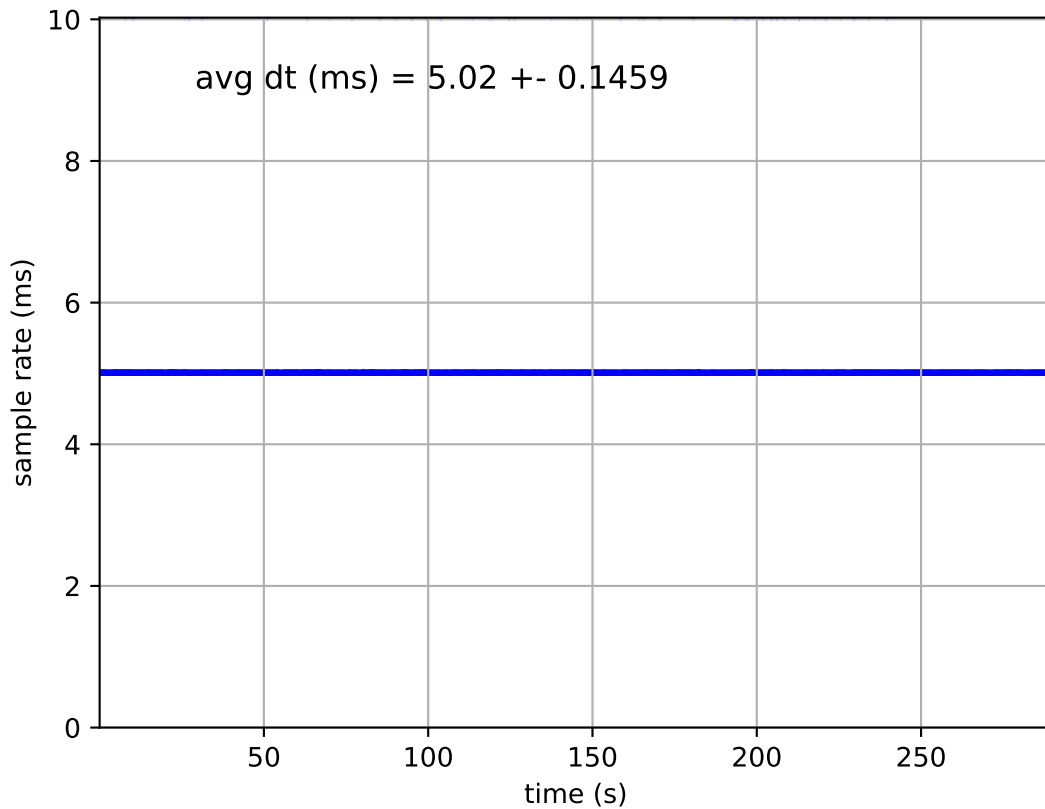
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

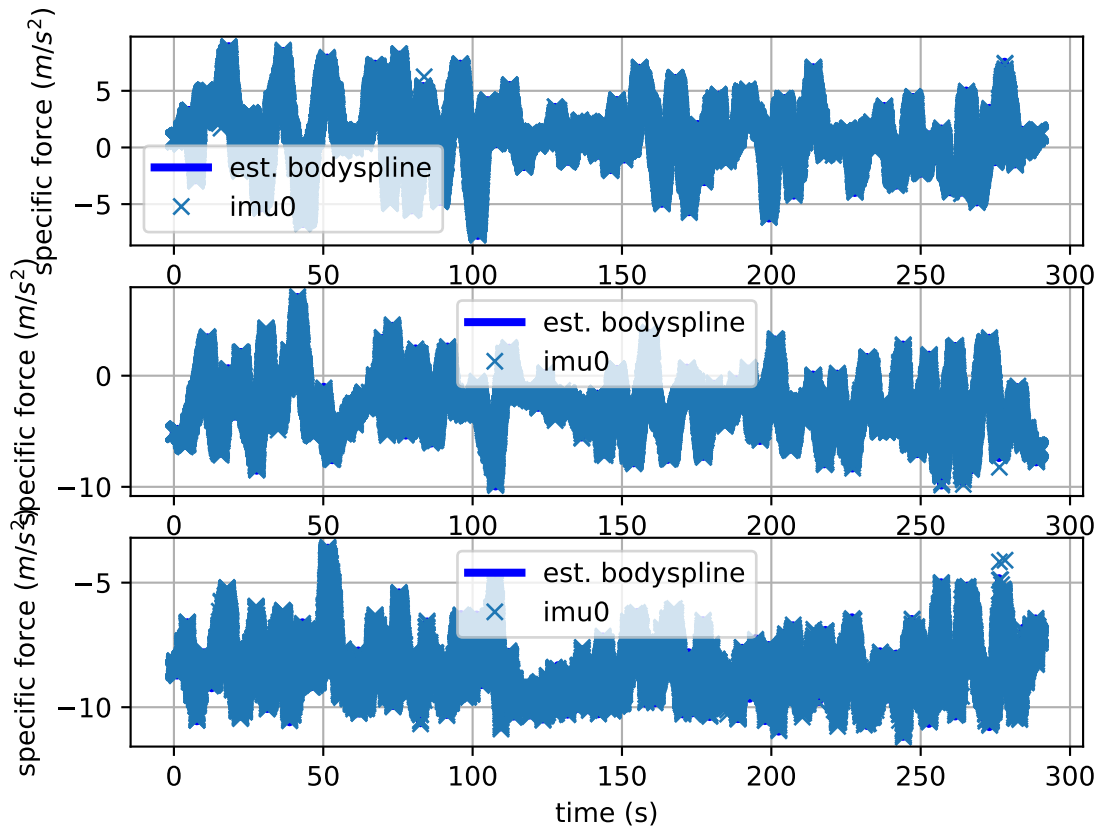
imu0: estimated poses



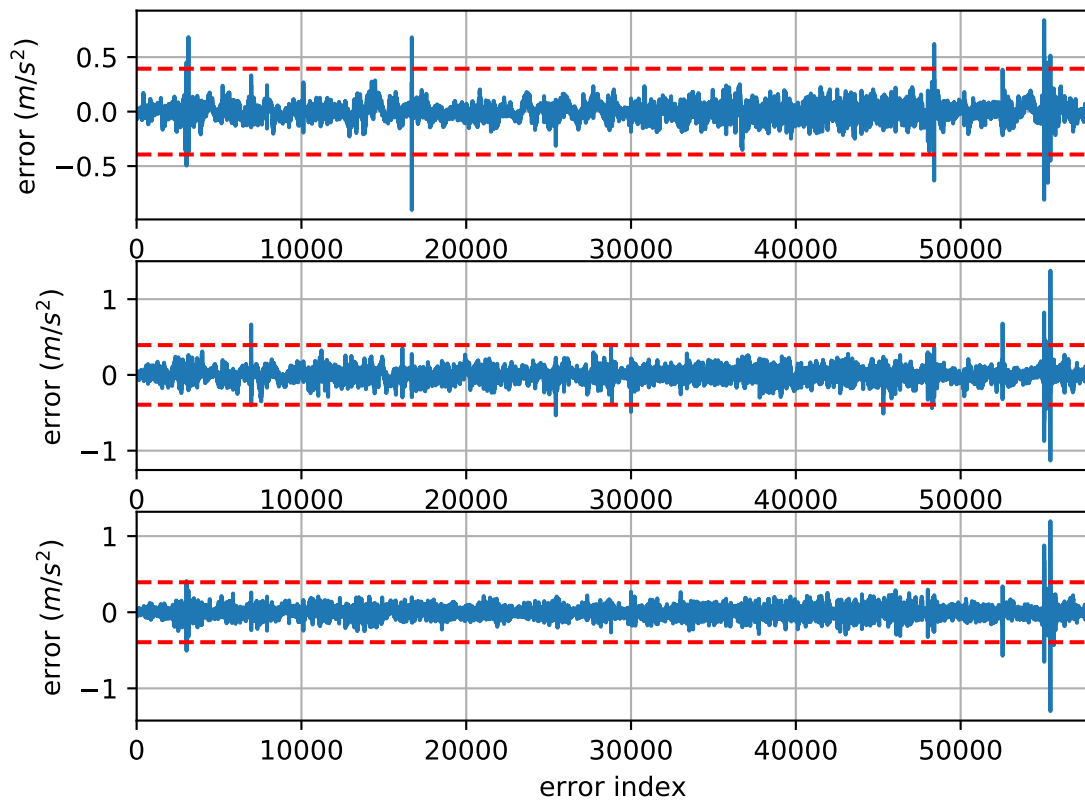
imu0: sample inertial rate



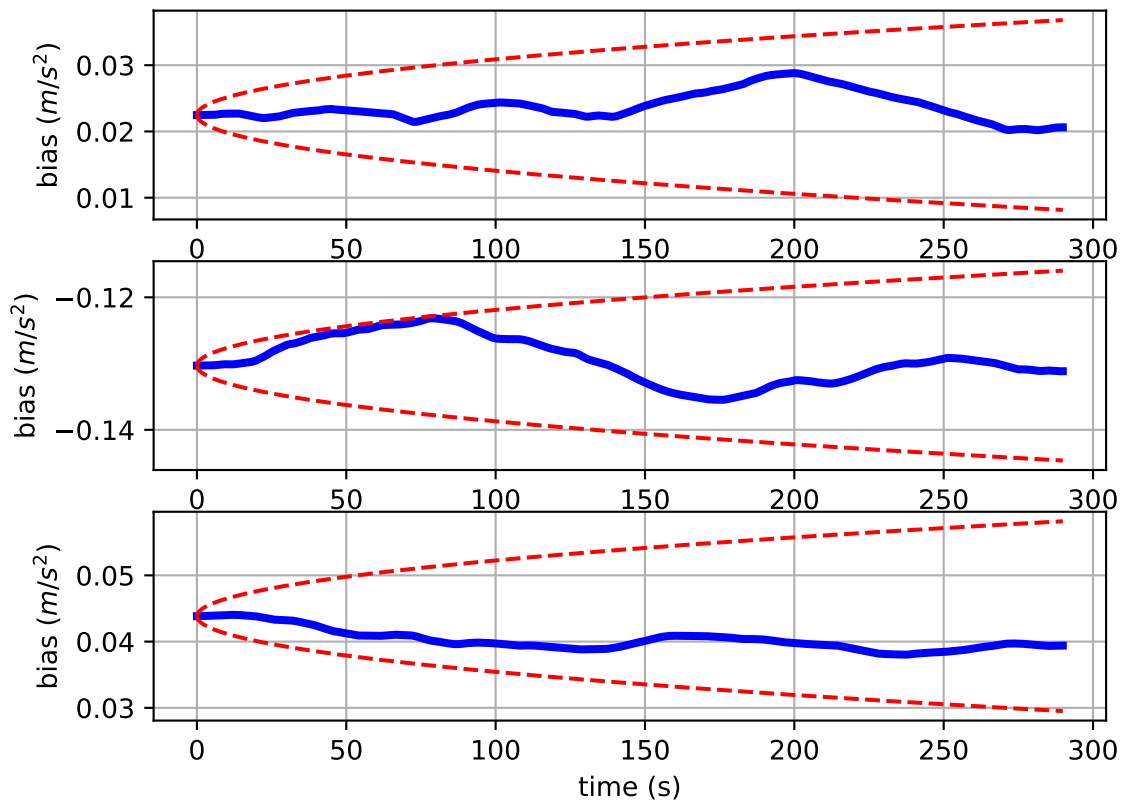
Comparison of predicted and measured specific force (imu0 frame)



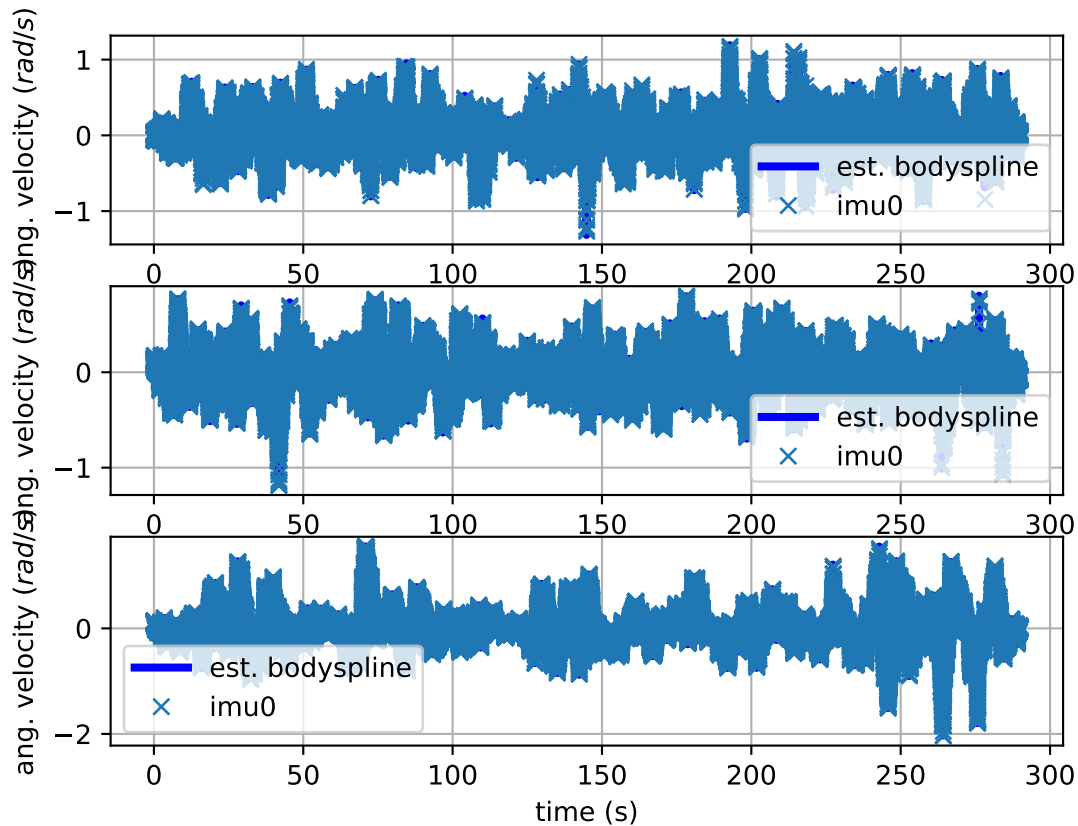
imu0: acceleration error



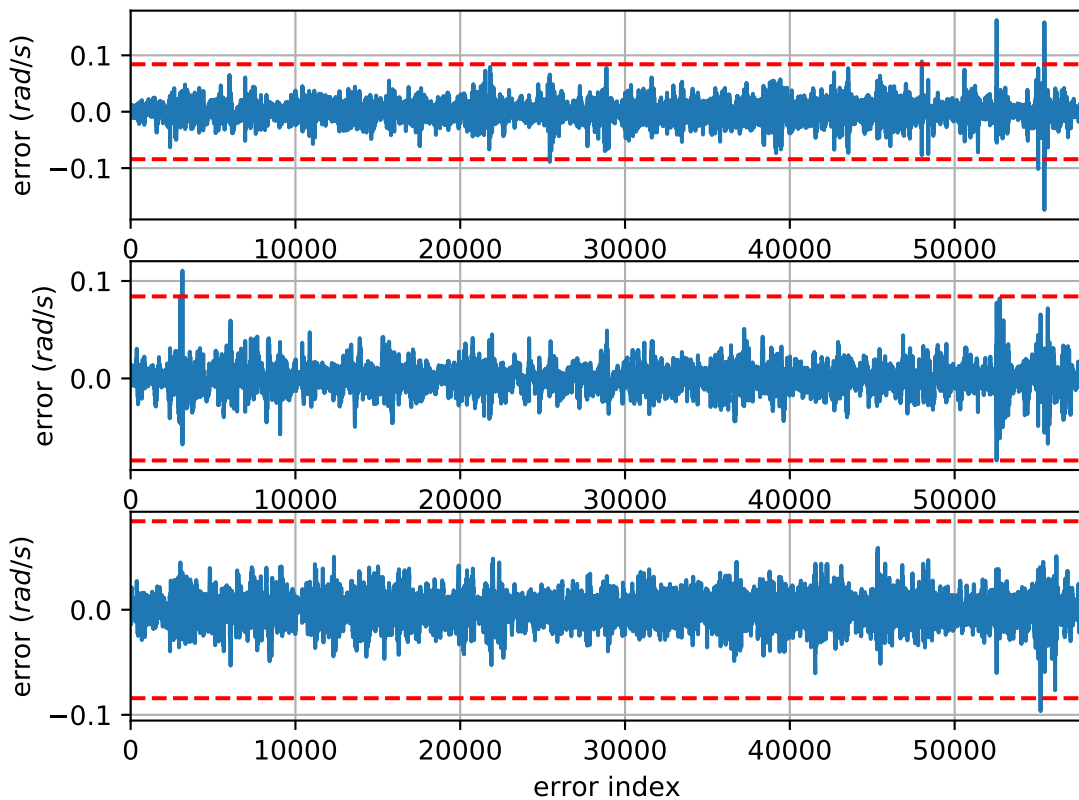
imu0: estimated accelerometer bias (imu frame)



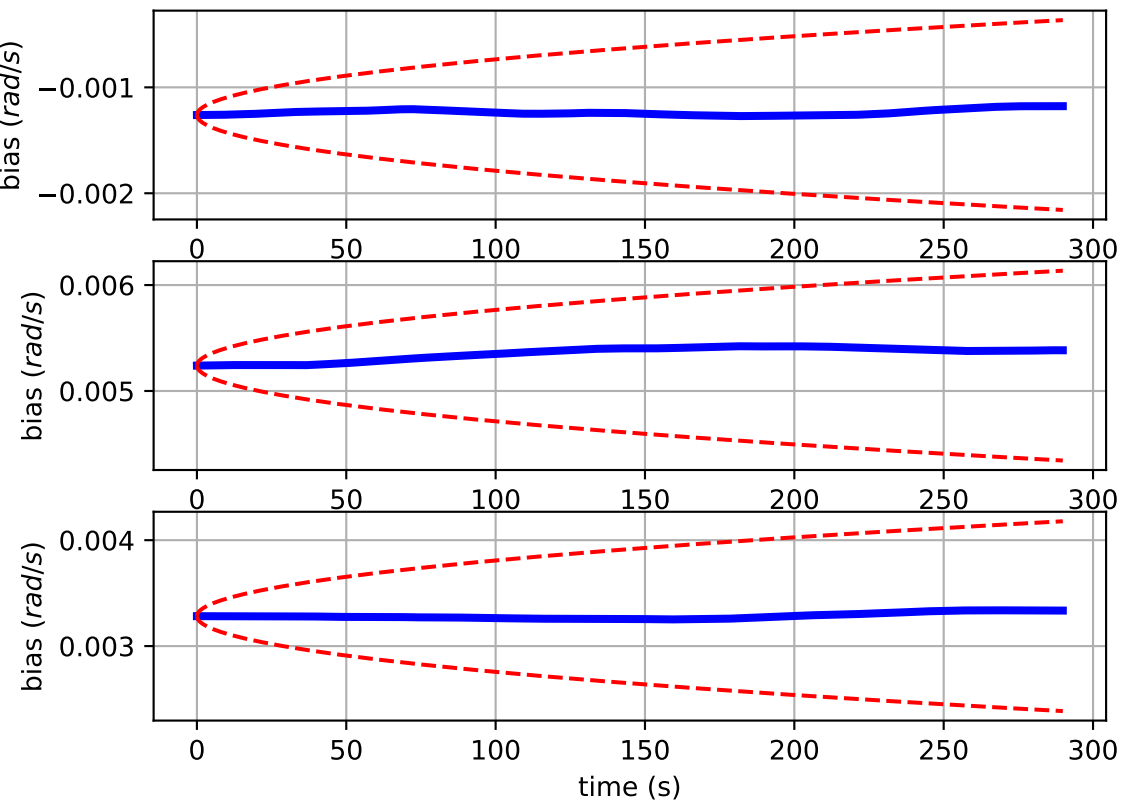
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

